BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric and Gas Service.

25-E-XXXX, 25-G-XXXX

January 2025

Prepared Testimony of:

Depreciation Panel

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I. INTRODUCTION AND PURPOSE OF TESTIMONY

Q. Would each member of the Depreciation Panel please state your name and business address?

A. My name is Qun Li. My business address is 4 Irving Place, New York, New York.

My name is Robert Benoit. My business address is 4 Irving Place, New York, New York

My name is Ned W. Allis. My business address is 207 Senate Avenue, Camp Hill, Pennsylvania.

My name is Melissa M. Howard. My business address is 207 Senate Avenue, Camp Hill, Pennsylvania.

Q. Ms. Li, by whom are you employed and in what capacity?

A. I am employed by Consolidated Edison Company of New York, Inc. ("Con Edison" or the "Company"). I supervise the book depreciation accounting functions as well as being the rate case lead on depreciation studies for the Company.

Q. Please briefly outline your educational background and business experience.

A. I received a Bachelor of Science degree from Wuhan Science and Technology University and a Master of

Accountancy from Bowling Green State University. I am a Certified Cost Engineer. I have been employed by Con Edison since March 2012 as a Senior Accountant. I am a member of the Society of Depreciation Professionals ("SDP") and have completed several training courses conducted by the SDP.

Q. Mr. Benoit, by whom are you employed and in what capacity?

A. I am Employed by Con Edison, where I am the Section Lead for the Property Records department. I am responsible for managing the Plant and Depreciation sections.

Q. Please briefly outline your educational background and business experience.

A. I received a Bachelor of Science degree in Economics from SUNY Old Westbury in Old Westbury, New York. I have worked in the Property Record department at Con Edison for eight years performing various closing and reporting functions, as well as supervising groups in the department. I have also worked in Gas Operations at Con Edison where I assisted in implementing the

work management system. Prior to Con Edison, I worked at National Grid for three years in the Fixed Asset department assisting with depreciation processing and closing.

Q. Mr. Allis, by whom are you employed and in what capacity?

A. I am employed by Gannett Fleming Valuation and Rate Consultants, LLC ("Gannett Fleming"), where I am Vice President. I am responsible for conducting depreciation, valuation and original cost studies, determining service life and net salvage estimates, conducting field reviews, presenting recommended depreciation rates to clients, and supporting such rates before state and federal regulatory agencies. I am also responsible for Gannett Fleming's proprietary depreciation software, training of depreciation staff, and the development of solutions for technical issues related to depreciation.

Q. Mr. Allis, please briefly outline your educational background and business experience.

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A. I received a Bachelor of Science degree in Mathematics from Lafayette College in Easton, Pennsylvania. I am a member of the SDP and am a past president of SDP. I am certified as a depreciation expert by the SDP, which has established national standards for certification via an examination that I passed in September 2011. I was re-certified as a depreciation professional in March 2017 and January 2022.
I became employed by Gannett Fleming in October 2006 as an Analyst. My duties included assembling basic data required for depreciation studies, conducting statistical analyses of service life and net salvage data, calculating annual and accrued depreciation, and assisting in preparing reports and testimony setting forth and defending the results of the studies. In March 2013, I was promoted to the position of Supervisor, Depreciation Studies. In March 2017, I was promoted to the position of Project Manager, Depreciation and Technical Development. In January 2019, I was promoted to my current position of Vice President.

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Q. Ms. Howard, by whom are you employed and in what capacity?

A. I am employed by Gannett Fleming, where I am an Assistant Project Manager specializing in depreciation. My primary duties are conducting depreciation studies for various types of utility companies and providing recommended service life and net salvage estimates for use in determining depreciation rates. I also provide support for these studies before regulatory agencies in rate proceedings.

Q. Ms. Howard, please briefly outline your educational background and business experience.

A. I received a Bachelor of Science degree in Civil Engineering, as well as a Master of Business Administration, from Temple University in Philadelphia, Pennsylvania. I also earned a Master of Science degree in Civil Engineering from Lehigh University in Bethlehem, Pennsylvania. I am a licensed Civil Engineer in the state of Pennsylvania,

and I earned certification as a depreciation professional ("CDP") from the SDP in 2019. I began employment with Gannett Fleming, Inc. (the parent company of Gannett Fleming) in July 2001 as a Civil Engineer-in-Training working in the Environmental Division. My primary responsibilities were performing hydrologic and hydraulic modeling for water resources projects including dam rehabilitation and construction projects. In 2007, I joined Gannett Fleming as a Depreciation Analyst. My duties included assembling data required for depreciation studies, conducting statistical analyses of service life and net salvage data, calculating annual and accrued depreciation and assisting in the preparation of reports of these analyses. In 2024, I was promoted to my current position of Assistant Project Manager.

Q. Have the members of the Depreciation Panel previously testified or submitted testimony in any utility commission on the subject of utility plant depreciation?

A. **(Li)** Yes. I have previously submitted testimony before the New York Public Service Commission ("NYPSC" or "Commission") in Cases 13-E-0030, 13-G-0031 and 13-S-0032 and 22-E-0064 and 22-G0065 for Con Edison and in Cases 24-E-0060 and 24-G-0061 for Con Edison's affiliate, Orange and Rockland Utilities, Inc. ("Orange and Rockland").

(Benoit) No.

(Allis) Yes. I have testified or submitted testimony on the subject of depreciation before the NYPSC in each Con Edison rate case commencing with Cases 13-E-0030, 13-G-0031 and 13-S-0032, as well as various Orange and Rockland rate cases, including Cases 24-E-0060 and 24-G-0061. I have also prepared depreciation studies and testified in cases for other New York State utilities and have testified on the subject of depreciation before the California Public Utilities Commission, Connecticut Public Utilities Regulatory Authority, District of Columbia Public Service Commission, Florida Public Service Commission, the Illinois Commerce Commission, Kansas Corporation

Commission, Maryland Public Service Commission, Massachusetts Department of Public Utilities, Maine Public Utilities Commission, Missouri Public Service Commission, Nevada Public Utilities Commission, New Hampshire Public Utilities Commission, New Jersey Board of Public Utilities, Rhode Island Public Utilities Commission, Tennessee Public Utility Commission, Virginia State Corporation Commission, and the Washington Utilities and Transportation Commission.

(Howard) Yes. I have previously submitted testimony in Cases 24-E-0060 and 24-G-0061 for Orange and Rockland.

Q. What is the purpose of your testimony in this proceeding?

A. The Depreciation Panel's testimony:

- Presents the Depreciation Study performed by Gannett Fleming for the Company's electric, gas and common plants;
- Presents annual depreciation accruals, as of December 31, 2023, based on the Company's

existing rates and the proposed depreciation rates;

- Identifies the Accumulated Provision for Depreciation recorded on the Company's books ("book reserve") as of December 31, 2023, the computed reserve (also referred to as the theoretical reserve or calculated accrued depreciation) based on existing depreciation factors, and the computed reserve based on the proposed depreciation factors for electric, gas and common plants;
- Presents the variations between the book and computed reserves based on existing rates and the proposed depreciation factors for electric, gas and common plants and the Company's proposal to recover the reserve deficiencies that exceed 10% of the theoretical reserve for electric and gas plant over 20 years; and
- Discusses the impact of New York State's law requiring decarbonization of the energy sector, *i.e.*, the Climate Leadership and Community

Protection Act ("CLCPA"), how this may impact depreciation for the electric and natural gas industries, and how the recommended service lives and reserve variation recovery for gas assets represent an initial step to recognize the potential impact of this legislation.

Q. Is the Depreciation Panel sponsoring any exhibits in these proceedings?

A. Yes. The Depreciation Study, which was prepared by Gannett Fleming and reviewed by Ms. Li and Mr. Benoit, is presented in the following exhibits prepared under our supervision and direction:

- Exhibit ___ (DP-1) entitled: "Consolidated Edison Company of New York, Inc., Depreciation Study, Electric, Gas and Common Plant as of December 31, 2023" ("Depreciation Study");
- Exhibit ___ (DP-2) entitled: "Consolidated Edison Company of New York, Inc., Electric, Gas and Common Plant, Summary of Annual Depreciation Rates as of December 31, 2023;"

- Exhibit ___ (DP-3) entitled: "Consolidated Edison Company of New York, Inc., Summary of the Computed Reserves for Depreciation as of December 31, 2023;"
- Exhibit ___ (DP-4) entitled: "Consolidated Edison Company of New York, Inc., Comparison of Annual Depreciation Expense Resulting from Various Scenarios for Gas Plant;"
- Exhibit ___ (DP-5) entitled: "Consolidated Edison Company of New York, Inc., Comparison of Proposed Survivor Curves to Average Service Lives and Remaining Lives from Life Cycle Analysis for Gas Plant Accounts;"
- Exhibit ___ (DP-6) entitled: "Consolidated Edison Company of New York, Inc., Composite Life Cycle Curve for Vintage 2023 Based on 60-R1 Survivor Curve and Gas System Retirements."

Q. Please summarize any changes to depreciation expense levels due to Gannett Fleming's depreciation recommendations.

A. As set forth in its direct testimony, the Company's Accounting Panel has computed, based on the depreciation rates we have supplied, that depreciation expense will increase in the Rate Year one by $131.5 million for electric plant (*i.e.*, from $1.62 billion to $1.75 billion) and by $93.2 million for gas plant (*i.e.*, from $488 million to $582 million.

Q. Do the Company's proposed depreciation rates account for the CLCPA?

A. Yes. As we have discussed in prior Company rate cases and in the Commission's Gas Planning Proceeding, the CLCPA will mean a need for higher depreciation for gas assets. Further, Company counsel has advised us that the Commission has determined that the CLCPA applies to utility rate cases. It is, therefore, appropriate to consider the potential impacts of New York State's carbon emissions goals in establishing the Company's depreciation rates and recovery of reserve deficiencies. In doing so, we recommend a decrease in the service lives for several of the longer-lived gas accounts by five years (when compared to the results

of service life analysis that does not consider the CLCPA's impact).

II. DEPRECIATION STUDY

Q. Please define the concept of depreciation.

A. The Electric FERC Uniform System of Accounts (18 C.F.R. 101, Definition 12) defines depreciation as follows:

> Depreciation, as applied to depreciable electric plant, means the loss in service value not restored by current maintenance, incurred in connection with the consumption or prospective retirement of electric plant in the course of service from causes which are known to be in current operation and against which the utility is not protected by insurance. Among the causes to be given consideration are wear and tear, decay, action of the elements, inadequacy, obsolescence, changes in the art, changes in demand and requirements of public authorities.

The Gas FERC Uniform System of accounts (18 C.F.R. 201, Definition 12) is similar:

> Depreciation, as applied to depreciable gas plant, means the loss in service value not restored by current maintenance, incurred in connection with the consumption or prospective retirement of gas plant in the course of service from causes which are known to be in current

operation and against which the utility is not protected by insurance. Among the causes to be given consideration are wear and tear, decay, action of the elements, inadequacy, obsolescence, changes in the art, changes in demand and requirements of public authorities, and, in the case of natural gas companies, the exhaustion of natural resources.

We note that both the Electric and Gas FERC Uniform Systems of Accounts specifically enumerate obsolescence, changes in demand, and requirements of public authorities as factors that should be considered in determining depreciation. We address these factors in this testimony and the Company will continue to assess them in future depreciation studies.

Q. In preparing the Depreciation Study, did you follow generally accepted practices in the field of depreciation?

A. Yes.

Q. Are the methods and procedures you used in the Depreciation Study consistent with the Company's past practices?

A. Yes. We used the same methods and procedures in the Depreciation Study as those used in previous depreciation studies conducted by the Company. We also note that the methods and procedures are consistent with those used in depreciation studies presented by other companies in rate proceedings before the Commission. The approach is to determine depreciation rates based on the straight line method, the broad group average service life procedure and the whole life technique. Consistent with the prior Depreciation Study and the Company's current depreciation rates, we used Iowa-type survivor curves to estimate average service lives.

Q. Please describe, generally, the presentation of the Depreciation Study in your exhibits.

A. Generally, the Company provides the following for the Depreciation Study: (1) background information; (2) results; and (3) graphs and tables.

Q. Please describe the presentation of the background information for the Depreciation Study.

A. The Depreciation Study, set forth in Exhibit ___ (DP-1), is presented in nine parts. Parts I through V provide background information for the Depreciation Study. Specifically, Part I, Introduction, presents the scope and basis for the Depreciation Study. Parts II through V include descriptions of the methods and procedures used for the estimation of survivor curves and net salvage and the calculation of annual depreciation and the theoretical reserve.

Q. Please describe the presentation of the results of the Depreciation Study.

A. Part VI, Results of Study, presents a description of the results and a summary of the depreciation calculations.

Q. Please describe the presentation of the graphs and tables for the Depreciation Study.

A. Parts VII through IX present graphs and tables that relate to the service life analyses, the net salvage analyses and the detailed depreciation calculations.

Q. Please continue.

A. The tables on pages VI-4 through VI-7 of Exhibit ___ (DP-1), present the estimated survivor curve, the net salvage percent, the original cost of plant and the book depreciation reserve as of December 31, 2023, and the calculated annual depreciation accrual and applicable depreciation rate for each plant account or subaccount. The section beginning on page VII-1 presents the results of the retirement rate analyses prepared as the historical bases for the service life estimates. The section beginning on page VIII-1 presents the results of the net salvage analysis. The section beginning on page IX-1 presents the depreciation calculations related to surviving original cost as of December 31, 2023.

Q. Please explain how you performed the Depreciation Study.

A. The Depreciation Study began with a planning meeting held on September 11, 2024. We conducted the Depreciation Study over the next several months and concluded with the report that is included as Exhibit ___(DP-1).

The Depreciation Study, included as Exhibit ___(DP-1), used the straight line, whole life method of depreciation, with the broad group average service life procedure. The annual depreciation is based on a method of depreciation accounting that seeks to distribute the service value (original cost of plant assets plus estimated costs of removal less estimated gross salvage at the time of retirement) over the estimated service life of each group of assets in a systematic and rational manner.

Q. How did you determine the recommended annual depreciation accrual rates?

A. This was done in two phases. In the first phase, we developed estimates of the service life and net salvage factors for each depreciable group, *i.e.*, each plant account or subaccount identified as having similar characteristics. In the second phase, we calculated the annual depreciation accrual rates using the applicable average service lives and net salvage factors.

Q. What part does the average service life play in the determination of depreciation rates?

A. The average service life is the period over which the original cost of plant will be depreciated. For example, with an average service life of 25 years, annual depreciation is 1/25, or 4%, of the original cost of the plant before taking into account the net salvage factor.

Q. What is the effect on annual depreciation expense of a change to an average service life?

A. The depreciation expense accrual varies inversely with its underlying average service life. In other words, a change that increases the average service life results in a lower annual depreciation rate and a lower annual depreciation expense. Conversely, a change that decreases the average service life results in a higher annual depreciation rate and a higher annual depreciation expense.

Q. What part does net salvage play in the determination of depreciation rates?

A. Depreciation is intended to recover the full costs of Company assets over the period of time they are in service. The full cost of an asset includes both the original cost when the asset was installed and the net salvage at the end of the asset's service life. Thus, in addition to providing for recovery of the original cost of plant over its estimated average service life, annual depreciation rates include an estimated net salvage factor. The purpose of this estimated net salvage factor is to reflect, over the life of the plant, the expected gross salvage value of plant less the expected cost of removal upon retirement. With few exceptions, plant assets typically result in negative net salvage upon retirement, which means that removal costs exceed gross salvage value. Gross salvage and removal cost values are netted, expressed as a percentage of the plant's original cost, and included in the annual depreciation rate. As a result, and in accordance with basic depreciation principles and the NYPSC's Uniform System of Accounts, the service value, including net salvage, of an asset

is allocated evenly over the estimated useful life of the asset.

Q. Please describe the first phase of the Depreciation Study in which you estimated the average service life and net salvage factors for each plant account or subaccount.

A. The service life and net salvage study consisted of compiling historical data from records related to Con Edison's plant; analyzing the data to obtain historical trends of survivor characteristics; obtaining supplementary information from management and operating personnel concerning practices and plans as they relate to plant operations; making visits to various sites to view the physical condition of facilities; and interpreting the data and information along with the average service lives and net salvage factors used by other electric and gas utilities to form judgments of average service lives and net salvage factors applicable to Con Edison's plant and equipment.

Q. For the preparation of the Depreciation Study, what is the significance of your visits to Con Edison facilities?

A. We made a field review of Con Edison property as part of the Depreciation Study during November 2024. Mr. Allis also conducted field reviews in December 2021, December 2018, October 2015 and November 2014 for the Company's previous depreciation studies. Depreciation studies should not be limited only to statistical analysis or visual comparisons of smoothed survivor curves based on actual mortality experience and standardized survivor curves. Field reviews, as well as discussions with operating and engineering personnel, are conducted to become familiar with Company operations and obtain an understanding of the function of the plant. Additionally, the reviews and discussions provide valuable information with respect to the reasons for past retirements and the expected causes of future retirements. We incorporated this knowledge, as well as information from other

discussions with management, in the interpretation and extrapolation of the statistical analyses.

Q. What historical data did you analyze to estimate average service lives?

A. We analyzed the Company's accounting entries that record plant asset transactions during the period 1938 through 2023. These transactions included additions, retirements, transfers and related balances.

Q. What method did you use to analyze the data?

A. We used the retirement rate method. This is the most appropriate method when retirement data cover a long period of time. The retirement rate method determines the average rates of retirement the Company has experienced during the period of time covered by the Depreciation Study. It is also the method Con Edison used in past depreciation studies and is overwhelmingly the predominant approach used in depreciation studies across the Country when aged data are available.

Q. Please describe how you used the retirement rate method to analyze the Company's service life data.

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A. We performed the retirement rate analysis for each discrete group of property, generally a particular plant account, in the Depreciation Study. For each property group, we used the retirement rate data to form life tables which, when plotted, show original survivor curves for that property group. Each original survivor curve represents the average survivor pattern experienced by the vintage groups during the experience band (the range of transaction years included in an original life table) and placement band (the range of vintage years included) studied. Because original survivor curves do not include the complete experience of most vintages of plant and because future survivor characteristics may be different from those experienced in the past, the experienced survivor patterns do not necessarily represent the property group's life characteristics. Therefore, interpretation of the original survivor curves is required for them to be valid considerations in estimating future service lives. Standard survivor

curves, such as the Iowa-type survivor curves, are used as a basis for these interpretations.

Q. What is an "Iowa-type survivor curve," and how can it be used to estimate the average service life characteristics for each property group?

A. Iowa-type survivor curves are a widely used group of survivor curves that contain the range of survivor characteristics usually experienced by utilities and other industrial companies. The Iowa curves were developed at the Iowa State College Engineering Experiment Station through an extensive process of observing and classifying the ages at which various types of property used by utilities and other industrial companies had been retired. Iowa curves were used in the Company's previous Depreciation Study and are the basis for the Company's current depreciation rates.

Iowa-type curves are used to smooth and extrapolate original survivor curves resulting from the retirement rate method. The Iowa curves can be used to determine the forecasted rates of retirement

based on the observed rates of retirement and the outlook for future retirements.

The estimated survivor curve designations for each depreciable property group indicate the average service life, the family within the Iowa system to which the property group belongs, and the relative height of the mode. The Iowa 50-R1.5 provides an example of these designations. The first designation indicates an average service life of fifty years. The second designation indicates a right-moded type curve, where the mode, or peak, occurs after average life for the R curve. The third designation indicates a relatively low height of 1.5 for the mode (possible modes for R type curves range from 1 to 5).

Q. Please provide an example of how the annual depreciation accrual rate for a particular plant account is presented in the Depreciation Study.

A. We will use electric plant Account 367.00, Underground Conductors and Devices, as an example because it is one of the largest depreciable accounts.

The Company used the retirement rate method to analyze the survivor characteristics of this property group. Aged plant accounting data, which track ages and historical costs of assets, were compiled from 1938 through 2023 and analyzed over a period that best represents the overall service life of this property. The life table for the 1938-2023 experience band is presented on pages VII-85 through VII-88 of Exhibit ___ (DP-1). The life table displays the retirement and surviving ratios of the aged plant data exposed to retirement by age interval. For example, page VII-85 shows $61,136,869 retired at age 0.5 years with $8,443,672,355 having been exposed to retirement. Consequently, the retirement ratio is 0.0072 ($61,136,869/8,443,672,355) and the survivor ratio is 0.9928 (1 – 0.0072). The life table, or original survivor curve, is plotted along with the estimated smooth survivor curve, the Iowa 55-R0.5, on page VII-84.

The calculation of the annual depreciation accrual and the theoretical reserve, or the expected

accumulated depreciation, related to the original cost of plant for Account 367.00, as of December 31, 2023, is presented on pages IX-60 through IX-63. The calculations are based on the 55-R0.5 survivor curve and 90% negative net salvage factor, and the attained age for each vintage. The tabulation sets forth the installation year, the original cost, average service life, calculated annual depreciation rate and accrual, average remaining life, and calculated accrued depreciation factor and amount, *i.e.*, the theoretical reserve ratio and theoretical reserve. The total annual accrual of $295,750,200 and theoretical reserve of $2,780,454,516 for the account are brought forward to the table on page VI-5. The reserve variation of negative $475,134,467 shown on page VI-5 is calculated by subtracting the $2,780,454,516 theoretical reserve from the book reserve amount of $2,305,320,049. A negative variation indicates that there is a book reserve deficiency for this account and that the depreciation rate likely needs to be higher than in the past.

Q. Please describe how you determined the proposed net salvage factors.

A. Consistent with well-established industry practices, we determined the net salvage factors by considering relevant information such as the results of historical net salvage analyses, the current net salvage rates, the Company's current practices with regard to net salvage, and the net salvage factors used by other electric and gas companies.

Q. Please describe the statistical net salvage analyses.

A. In the statistical net salvage analyses, net salvage is expressed as a percentage of the book cost of plant retired by calendar year. This method provides a statistical basis for the level of net salvage, as a percentage of original cost, that can be expected to occur in the future.

Q. Please explain why net salvage is analyzed as a percentage of retirements.

A. The net salvage percentages recommended in the study are estimates of the future net salvage costs related to assets currently in service. Net salvage is

estimated as a percentage of original cost or, equivalently, as a percentage of future retirements. In order to estimate a net salvage percentage to apply to the current plant in service, the net salvage analysis expresses historical net salvage as a percentage of historical retirements. In doing so, the analysis recognizes that net salvage in a given year is a function of the number of assets retired in that year.

For example, if 100,000 poles were retired in a given year, the net salvage would be greater than if 1,000 poles were retired and, all else equal, would be greater by a factor of a hundred. As a result, the net salvage analyses expressed as a percentage of retirement provides a consistent basis to analyze historical net salvage percentages to be applied to current plant in service and, therefore, provide an estimate of future net salvage.

Q. Are the Company's net salvage analyses and approach to reflecting net salvage in depreciation rates consistent with recommendations of the National

Association of Regulatory Utility Commissioners ("NARUC") and other depreciation authorities?

A. Yes. NARUC states the following concerning the analysis used by the Company (*See* Public Utility Depreciation Practices, National Association of Regulatory Commissioners, 1994, p. 159):

> Once the source of information is established, the analysis of data can commence to determine the past relationship of net salvage to retirements, i.e., net salvage as a percent of plant retired for each of the depreciation categories being studied. Net salvage can be directly analyzed as a percent of retirements. However, in order to obtain a clear understanding of the composition of net salvage and the forces that cause it to change from year to year, generally it is best to analyze gross salvage and cost of removal as a percent of retirements. In making this analysis it is common to look at data for bands of years, such as 1988-93 ,1989-1994, 1990-1995, etc.

Q. Are the methods you used in the Depreciation Study for the net salvage analysis widely accepted in the industry?

A. Yes. The net salvage analysis used in the Depreciation Study is the predominant approach in the

utility industry. In the vast majority of jurisdictions, including New York State, a portion of depreciation expense is based on the prospective recovery of future net salvage over the service life of the assets, and the net salvage factors are estimated using the same methods used in the Depreciation Study. This approach is consistent with the NYPSC Uniform System of Accounts, the ratemaking practices of most other state regulatory commissions, and the ratemaking approach of the FERC.

Q. Has the Company's method for analyzing net salvage been long used in New York?

A. Yes. The Commission most recently affirmed the use of the net salvage analysis in Cases 23-E-0418 and 23-G-0419. Department of Public Service ("DPS") Staff has also used the method of analyzing net salvage as a percentage of retirements in most cases in which Gannett Fleming has been involved.

Q. What are you proposing for gas assets expected to be retired as a result of the Company's leak-prone pipe replacement program?

A. For the Depreciation Study, the Company's gas mains and services accounts were segregated into the portion that will be retired through the Company's pipe replacement program and all other mains and services. Con Edison currently plans to complete this program by the end of 2040. Accordingly, our proposal for the assets that are planned to be replaced, which consist primarily of older cast iron and bare steel pipe, is to recover the remaining costs by 2040. This is the same approach as used in Con Edison's most recent rate case (Case 22-G-0065), as well as for other New York State utilities.

Q. In addition to the pipe replacement program, are you proposing any other notable changes in this case to the service life estimates for gas assets?

A. Yes. We propose to shorten the service lives for several of the longer-lived gas accounts (*i.e.*, mains, measuring and regulating equipment, meters, meter installations, house regulators and house regulator installations) from what the Company has historically experienced due to the impacts of the CLCPA and the

clean energy transition underway in New York State. For most of these accounts, our recommendation is to shorten the average service lives by five years when compared to the lives that would result from a depreciation study that does not consider the effects of the CLCPA. These impacts and our proposal will be discussed in more detail in the next section.

III. CLIMATE LEADERSHIP AND COMMUNITY PROTECTION ACT

Q. Does the Company's rate filing reflect potential impacts of the CLCPA?

A. Yes, although it does so moderately. The Company's proposal recognizes that higher depreciation, which will be necessary due to the CLCPA, has short-term impacts on customers, as well as long-term impacts on both the Company and its future customers. As we discuss further, the recommendation for moderate reductions in service lives is consistent with the concept of gradualism and mitigates short-term impacts on customers.

The Company's proposal also recognizes that there is a need to begin to take action on this issue, and

to do so with a sense of urgency. While we recognize concerns related to the cost impact for current customers, the result of continuing to defer recognition of the reality of the CLCPA also has significant cost impacts for customers, particularly the Company's future customers. The Company has a large and growing reserve deficiency, which increases customer rates. This is due to the need to recover these costs through additional depreciation and because the reserve deficiency results in a higher rate base and corresponding return on rate base.

The deferral of depreciation expense already impacts the Company's current customers, but the impact will grow in the coming years, particularly if the necessary increases in depreciation continue to be deferred.

Q. Is this the first case in which the Company or its affiliates have proposed to address the impacts of the CLCPA on depreciation?

A. No. In its previous rate case, Case 22-G-0065, the Company proposed to shorten service lives to address

the impacts of the CLCPA. The Company's affiliate, Orange and Rockland, also made similar proposals in Cases 24-G-0061 and 21-G-073. We also note that other New York State utilities have made similar proposals.

The Company has also provided additional analyses over the past five years. In Cases 19-E-0065 and 19-G-0066, Con Edison filed a depreciation study that considered these impacts, as well as other potential ratemaking approaches to address declining demand and customer counts for the gas industry. In November 2022 under the Gas Planning Proceeding (Case 20-G-0131), both Con Edison and Orange and Rockland filed depreciation studies further assessing the long-term impacts of different depreciation approaches. These studies showed that methods or approaches that produced higher depreciation today will have significant benefits in appropriately reduced rates for future customers.

The Company and Orange and Rockland have also prepared a Gas System Long-Term Plan ("GSLTP"), including an initial plan filed with the Commission in

Case 23-G-0147 on May 31, 2023 and a final plan on November 29, 2023 in the same case. DPS Staff also filed a report on the issue in Case 23-G-0147 on December 12, 2023. Each of these analyses forecast declines in both gas demand and the number of customers taking natural gas service.

Q. To date, has the Commission issued a decision reflecting the results of these studies?

A. No. Con Edison first provided a study on this issue five years ago. Additionally, it has been more than two years since the filing of the analyses in the Gas Planning Proceeding. This is not a sufficient response to the significant challenges faced by the gas industry. Particularly, we see a broad consensus in the industry that the capital recovery of gas assets will be impacted by the CLCPA.

Q. Is there an industry expectation that the CLCPA will impact gas assets?

A. Yes. For example, both the Climate Action Council's Scoping Plan and the Commission's Gas Planning Proceeding have provided analyses and potential

pathways for decarbonizing electric and gas systems, as well as the economy as a whole. Based on our involvement and review of these analyses, there is a high probability that decarbonizing will result in significant changes to the electric and gas industries. Particularly for the gas industry, depreciation will need to increase to recover capital at a faster rate than in the past.

The studies and analyses provided in rate cases and the Gas Planning Proceeding have consistently shown that continuing a business-as-usual approach is unsustainable and will cause future gas rates to become increasingly unaffordable for many customers. There needs to be action to increase depreciation to align with the realities of the CLCPA. Failing to do so will not only be inequitable to future customers but also will threaten the ability of the State to meet its emissions goals.

Q. Please describe further how the GSLTP will impact the Company's assets.

A. The long-term plan has three primary scenarios -- a Reference Pathway, Hybrid Pathway and Deep Electrification Pathway. The Reference Pathway does not achieve CLCPA goals and is instead based on business as usual. The other two pathways do meet CLCPA emissions goals. The Hybrid is alternatively based on low-carbon fuels ("LCFs") and a smaller, less utilized gas system. The Deep Electrification Pathway envisions widespread electrification of most gas customers. Under both the Hybrid and Deep Electrification Pathways, gas throughput and the gas system footprint decline materially. For example, for the Hybrid Pathway, gas throughput would decline 52% for Con Edison and the overall footprint of Con Edison's gas system would decline by 50%. These are significant changes that will materially reduce both service lives (because assets will be retired as the system is downsized) and utilization of the Company's assets. Importantly, this will have profound impacts on both the gas system and the electric system.

Q. What conclusions can be drawn from these analyses?

A. Each of the pathways we have reviewed (both in New York State and in our work in other jurisdictions) anticipate electrification of many current energy uses, most notably transportation (through electric vehicles) and space and water heating (primarily through electric heat pumps). While the exact degree of electrification may vary in different analyses or scenarios, there are several common themes:

1. Widespread vehicle electrification will increase peak electric loads, not only the intensity of those peaks but also their geographic and temporal location within the system.

2. Large-scale renewable deployments will require significant upgrades to both electric distribution and transmission systems in order to incorporate intermittent generation located at different points throughout the system.

3. Widespread electrification of heating will increase peak loads, including changes to a winter peak, which is expected to occur in the late 2030s or

early 2040s. This will require rebuilding large parts of the electric system in order to meet the demands of widespread electrification.

4. Gas throughput and the footprint of the gas system will decline materially by 2050, with the most pronounced declines occurring concurrent with increases in electric demand in the late 2030s or early 2040s. Many gas customers may also leave the gas system and fully electrify. Under the more moderate Hybrid Pathway, the gas system is expected to be half the size it is today.

5. High electrification scenarios tend to have higher costs to customers than more moderate electrification scenarios in which the gas system is more widely utilized, particularly for peak heating. This is true not only for gas rates, but also for electric rates and customer-side costs (*e.g.*, converting to electric heat pumps, incremental costs for electric vehicles)

6. Capital will need to be recovered at a faster rate than in the past to address these changes and produce equitable rates that do not disproportionally harm future customers, particularly low-income customers.

Q. Has the Company proposed to address this issue in any recent rate cases?

A. Yes. In Case 22-G-0065, Con Edison proposed to shorten service lives for several accounts by as many as 10 years. In Case 24—G-0061, its affiliate, Orange and Rockland, proposed to shorten service lives for several accounts by 15 years.

Q. What is the Company proposing in this case?

A. The Company's proposal is to take a moderate step to incorporate the impacts of the CLCPA for gas assets. Specifically, we propose to shorten the service lives of several accounts by five years to incorporate the future outlook for the service lives of associated assets set forth in the GSLTP. The Company's current service life estimates for assets such as gas mains are as long as 90 years. In both the Hybrid and Deep

Electrification Pathways, service lives will be significantly shorter based on the downsizing of the gas system.

Q. Is this proposal the only one you have considered or analyzed when preparing the Depreciation Study?

A. No. We have also considered several other scenarios, most of which generally align with either previous proposals or the analyses included in the Gas Planning Depreciation Study filed in Case 20-G-0131. That study also analyzed depreciation rates that recover all costs by 2050 and Units of Production scenarios.

The table included in Exhibit ___ (DP-4) is a summary of different scenarios considered and compares each to the depreciation expense that results from the Company's current depreciation rates. The first scenario, labeled "Base Case," shows the results of a study if depreciation is determined in a traditional way with service life estimates more consistent with the Company's historical service life experience and using the straight line method. The other scenarios in the table, which show the results of different

approaches incorporating CLCPA impacts, include three scenarios with shorter service lives, with names beginning "3-Year Shorter," "5-Year Shorter" and "10-Year Shorter." As the names imply, these scenarios incorporate varying degrees of shorter service lives for certain accounts. We note that in each scenario the lives are not all uniformly shortened by the same amount, *e.g.*, 10 years shorter. However, for the larger accounts, such as mains and services, the degree to which lives have been shortened corresponds to the scenario name. Additionally, the table shows the results of using the Equal Life Group procedure as well as a scenario in which all costs are recovered by 2050. Finally, we show Units of Production ("UoP") scenarios aligned with the Hybrid and Deep Electrification Pathways.

Q. Does the table included in Exhibit ___ (DP-4) also reflect the impact of each scenario on the amortization of reserve deficiencies?

A. Yes. All else equal, shorter service lives result in a higher theoretical reserve which, in turn, results

in larger reserve deficiencies. The Reserve Variation Amortization row in the table shows the result of amortizing the reserve variation that exceeds 10% of the theoretical reserve over a 20-year period for each scenario. Reserve Variations are discussed in more detail in the next section.

Q. Given the results of these scenarios, what does the Company propose?

A. In consideration of the modeled scenarios shown above, as well as the factors that will potentially impact the service lives of the Company's assets, we recommend the 5-Year Shorter Lives scenario shown in the table, which includes a 20-year amortization of the reserve variation for gas plant that exceeds 10% of the theoretical reserve. We also believe the Units of Production Scenarios are reasonable approaches because they most closely align depreciation with the utilization of the Company's assets, as demonstrated in the Gas Planning Depreciation Studies, while mitigating rate impacts on future customers. However, the UoP is a change in depreciation method, dependent

on the pathway to 2050, and may perhaps be better addressed in a generic proceeding like the Gas Planning Proceeding. Therefore, our proposal in this case is based on the straight line method of depreciation currently in use and focuses on the impacts on service lives.

Q. Are any of the scenarios shown in the table included in Exhibit ___ (DP-4) accelerated depreciation?

A. No. The shortening of service life estimates that we are proposing in the Depreciation Study, and as shown in the scenarios in the table, is not accelerated depreciation. Accelerated depreciation describes methods of depreciation in which depreciation is higher in the earlier years and lower in the later years of an asset's life (when compared to the straight line method). None of the scenarios shown in the table use accelerated depreciation. Each uses the straight line method, and the only differences are assumptions about the service lives of the Company's gas assets.

Q. How does your proposal compare to the other scenarios set forth in the table?

A. While the 5-Year Shorter Service Lives scenario results in higher depreciation than the current depreciation rates or the Historical Experience scenarios, it is lower than either of the Units of Production Scenarios. The UoP scenarios result in the most equitable allocation of costs if throughput declines are consistent with either pathway modeled above. Given the outlook for the future, the most equitable level of depreciation is likely higher. As a result, it is our judgment that the 5-Year Shorter Service Lives scenario moves depreciation closer to where it should be, albeit gradually.

Q. In addition to qualitative expectations about the future, what statistical support do you have for the 5-year Shorter Service Lives scenario?

A. In addition to incorporating information from various pathways reports, we have also analyzed the direct impact of downsizing the gas system on service lives. To do so, we have used a technique called Life Cycle

Analysis to determine the combined impact of normal retirements and retirements that will occur as the gas system is geographically downsized.

For this analysis, we used the expectation of a 30% decline in the size of the gas system (as measured by the number of customers) based on the Hybrid Pathway. This means that by 2050 we would expect over half of the Company's assets to be retired when no longer providing service to (by 2050, fully electrified) customers. Using Life Cycle Analysis, we can combine the probabilities of retirement resulting from our survivor curve estimates with the retirements that will occur as the system is downsized.

Q. Please explain how these probabilities of retirement are combined to develop an overall Life Cycle curve.

A. Consider, as an analogy, a scenario in which there is a 10% chance it will snow today and a 10% chance it will rain. Assume these probabilities are mutually exclusive (meaning that it will exclusively rain, snow or do neither but will not both snow and rain), we can

combine them by multiplying the probabilities of surviving.

There is a 90% chance that it will not snow today. If it does not snow, then there is a 90% chance it also will not rain. The total probability of it neither snowing nor raining, then, is the 90% probability of no snow multiplied by the 90% probability of no rain, or 81%. That is, there is an 81% chance that it will neither snow nor rain today.

The same concept can be applied to utility plant and survivor ratios for each year. The combined probability of survival from different causes of retirement (*e.g.*, normal wear and tear as well as obsolescence due to the CLCPA) for a given age is similar to the calculation of the probability that it will neither rain nor snow today. If there is a 1% chance of retirement due to normal wear and tear and a 1% chance of retirement due to obsolescence, then the combined probability of surviving to the next age is 98.01% (or 99% x 99%). The figure set forth in Exhibit ___ (DP-5) shows the results of applying the

same mathematical techniques to the combined retirements resulting from downsizing the gas system by 52% by 2050 with the 60-R1 survivor curve estimate for the 2023 vintage for Account 380.10, Services.

The solid black line in the figure shows the 60-R1 survivor curve estimate. The dashed line shows the result of combining this estimate with retirements from downsizing the gas system. Once these retirement ratios are combined and a composite Life Cycle curve is developed, average service lives and remaining lives can be calculated using the same methods as for an Iowa survivor curve. More precisely, the average service life is the area under the full survivor curve and the remaining life is the area from a given age to the end of the curve, divided by the percent surviving at that age.

For this vintage, the impact of downsizing the gas system over the next three decades is to reduce the average service life (which is equal to the area under the composite survivor curve shown in the figure) from 60 to 46.62 years.

By applying the same technique to each vintage within the account and incorporating the current balances, we can then calculate an overall average service life and average remaining life for the entire account. For services, the result is shortening the 60-year average service life estimate from the 60-R1 survivor curve to a 50.04 year average service life, meaning that the Hybrid Pathway will result in a reduction in average service life of approximately 10 years for this account.

Q. Please summarize this analysis for each account for which you propose a shorter service life.

A. The table included in Exhibit ___ (DP-6) shows the results of this analysis for each account for which we propose an adjustment due to the CLCPA. As the table shows, this analysis is directionally consistent with the 5-year Shorter Lives scenario we propose. For example, the overall average service life based on our proposal for these accounts is 66.68 years, which is longer than the 47.49-year average service life resulting from the composite life cycle curves

calculated for each account. This demonstrates that, while our recommendation shortens service lives, the impact of the CLCPA will have a larger impact than our proposal. We consider our recommendation a very gradual change, given there is limited time to begin to recover the costs of gas assets.

Another way of considering this analysis is that the Hybrid Pathway will result in shortening the lives of the major gas assets by many years, on average. Our proposal recognizes some gradual shortening of service lives that will result from retirements based on the Hybrid Pathway. Importantly, the Deep Electrification Pathway would result in even more incremental retirements than the Hybrid Pathway and would, therefore, result in even shorter service lives.

Q. Are there risks to not reflecting the impact of the CLCPA in the Company's gas depreciation approach?

A. Yes. There are several risks. The first risk is the potential for future customers to bear the impact of cost recovery associated with infrastructure that is

not used to meet their energy needs, causing intergenerational inequity. If depreciation rates are too low today, future customers will have to pay an excessive share of the cost of the Company's assets while a transition to other energy sources takes place. Further, there is the risk that customers will leave the system as they electrify their energy usage, which would push additional costs to the future customers that remain. These risks are related. If depreciation is higher in the future and customers have left the system, there will be fewer customers to pay the remaining costs of the Company's assets, further compounding the intergenerational inequity resulting from depreciation rates being too low today.

Finally, there are additional equity concerns because the customers who remain may be disproportionately low- and moderate-income customers who are not able to electrify their energy usage as easily as customers with more resources. That is, if the recognition of the CLCPA's impact on depreciation is deferred to future cases, there is a risk that the

customers who bear a disproportionate share of the costs of the CLCPA will be those least able to afford these costs.

Q. Are there similar risks if depreciation rates are set too high today?

A. No. The risks are not symmetric. If depreciation rates are set too high today – and if in the future assets live longer and customers have not left the system – then depreciation rates will be adjusted through the amortization of reserve variations. Because the Company has reserve variations for both electric and gas service, this likely means that there would be smaller reserve variations to address in the future. Customers will pay lower costs for depreciation in the future. Additionally, rate base will be lower than it otherwise would have been, further reducing costs for customers (as well as Company earnings). In contrast, if depreciation rates are too low today – and if customers electrify and leave the gas system – then the impact on future customers will be much greater because there will

likely be fewer customers to pay the remaining capital costs. They will also have to pay a higher return on rate base, further compounding the issue. Finally, they will likely have to bear the costs of assets that are retired without being fully recovered, which is also inequitable.

For these reasons, the risks resulting from the CLCPA's goals are most appropriately dealt with by incorporating the potential for shorter asset lives into depreciation rates today. The sooner the Commission incorporates these factors, the lower the risk to future customers, the lower the potential for rate shock in the future, and the lower the total cost to customers over time. Deferring these decisions will both increase the risk of dramatic impacts on future rates and will cost customers, particularly low- and moderate-income customers, more in the long run.

Q. Will the Company continue to assess the impact of the CLCPA in future depreciation studies?

A. Yes. As the pathways to achieving the CLCPA's requirements further develop, the Company will assess their impacts on the depreciable lives of its gas assets. This stepped approach recognizes the CLCPA and the reality that the pathways to reach it are likely to affect the gas business, while maintaining flexibility. Further, this approach will allow the Company to adapt to future trends, regulations and technological advances. The Company's proposal in this case is reasonable given the current information and analyses available, but with additional information in future studies there may be a need to modify the approach or recommended depreciation rates.

Q. How has the impact of the CLCPA been considered for electric assets?

A. While we have not made explicit adjustments to electric service lives, we do believe it is important to begin considering electric industry impacts. The CLCPA will affect electric assets in a variety of ways. The change in electric generation from fossil power plants to renewables will cause changes in the

transmission system, as substations and transmission lines will have to be added, upgraded or replaced to incorporate energy from different sources in various locations. There will potentially be similar changes to the distribution system due to both the additions of distributed energy resources ("DERs") and changes in the load due to electrification.

As discussed previously, electrification will result in not only higher peak demand but also winter peaks. Assets such as transformers, circuit breakers and conductors will need to be replaced in order to meet higher loads and provide higher capacity. Reliability and resilience will also be important, particularly during winter peaks, which will also result in replacement of assets at a higher rate than in the past. Investments to upgrade the electric grid will have to incorporate the impact of a warming climate and the potential for more frequent and severe weather events. New technologies, which often have shorter service lives than legacy assets, will also be added to the electric grid.

At this time, we have not proposed an adjustment to average service lives due to these factors. However, the CLCPA is likely to result in significant investments in the electric system, and these investments may result in the replacement of a variety of assets. Accordingly, at a minimum we believe that any increases in service lives should be limited due to the transformational changes to the grid that will occur over the next two decades.

IV. TEST OF THE BOOK RESERVES

Q. What are the amounts of the variations between the book reserves and theoretical reserves that you mentioned earlier in your testimony?

A. For electric plant, the amounts we will address are summarized on Exhibit ___ (DP-3). This exhibit indicates that for total electric plant as of December 31, 2023, the Accumulated Provision for Depreciation per books, or book reserve, amounted to approximately $9.20 billion. The computed or

theoretical reserve based on existing rates was calculated on the average service lives, net salvage percentages and life tables currently in use by the Company and amounted to approximately $11.73 billion. The computed reserve recommended in the Depreciation Study amounted to approximately $11.89 billion.

This exhibit also indicates that the book reserve is approximately $2.54 billion, or 21.61 percent less than the computed reserve based upon existing rates. It is approximately $2.69 billion, or 22.62 percent less than the computed reserve based upon the rates recommended in the Depreciation Study.

Q. Please continue with gas plant.

A. For gas plant, the amounts we will address are summarized on Exhibit ___ (DP-3). This exhibit indicates that for total gas plant as of December 31, 2023, the book reserve amounted to approximately $2.45 billion. The theoretical reserve based on existing rates was calculated on the average service lives, net

salvage percentages and life tables currently in use by the Company and amounted to approximately $3.06 billion. The computed reserve recommended in the Depreciation Study amounted to approximately $3.29 billion.

This exhibit also indicates that the book reserve is approximately $611.50 million, or 19.97 percent less than the computed reserve based upon existing rates. It is approximately $840.03 million, or 25.53 percent less than the computed reserve based upon the proposed rates.

Q. Please continue with common plant.

A. For common plant, the amounts we will address are summarized in Exhibit ___ (DP-3). This exhibit indicates that for total common plant as of December 31, 2023, the book reserve amounted to approximately $1.55 billion. The theoretical reserve based on existing rates was calculated on the average service lives, net salvage percentages and life tables

currently in use by the Company and amounted to approximately $1.64 billion. The computed reserve based on the recommendations in the Depreciation Study amounted to approximately $1.64 billion.

This exhibit also indicates that the book reserve is approximately $84.96 million, or 5.19 percent less than the computed reserve based upon existing rates. The book reserve is approximately $84.84 million, or 5.18 percent less than the computed reserve based upon the rates recommended in the Depreciation Study.

Q. Do you have a recommendation regarding the reserve variations?

A. Yes. For common plant, we recommend no action be taken related to the reserve variations, at the levels indicated, at this time. The NYPSC's customary practice has been to take no remedial action when the book reserve varies from the theoretical reserve by less than 10% (plus or minus). The variation for common plant falls within that range.

For electric and gas plant, we recommend that the reserve variation above 10% of the theoretical reserve amount be recovered over a 20-year period. The estimated reserve variation for gas plant differs from the theoretical reserve by more than 10%, which indicates a need for remedial action. We recommend a 20-year recovery of these costs, which is a period that has been used in previous studies.

Q. Does that conclude your direct testimony?

A. Yes, it does.